

Executive Memo — Customer Churn

Customer Churn Analysis — Executive Memo

1. Executive Summary

- Out of **7,043 customers**, **1,869 (26.5%)** have churned — meaningfully above any tolerable steady-state.
- A tuned XGBoost model identifies churn risk with **PR-AUC 0.250 / ROC-AUC 0.804**, capturing **41.9% of churners by targeting only the top 10%** of accounts (lift = **4.19x**).
- Three actionable segments concentrate most of the avoidable churn; targeted outreach on the top-10% risk list could save an estimated **\$42,818/yr** at a conservative 30% save-rate.

2. EDA Key Findings

- Class balance: 73.5% retained vs 26.5% churned — moderate imbalance, handled with `scale_pos_weight`.
- Tenure is the dominant signal: short-tenure (0–12 mo) cohorts churn at multiples of the long-tenure cohort (see `cohort_churn.png`).
- Contract type, payment method, and internet service are the most discriminating categorical drivers (`segment_churn.png`); month-to-month + electronic-check + fiber-optic is the classic high-risk profile.
- Detailed prose narrative: `eda_narrative.md`.

3. Model Performance (held-out time-aware test set, n=1,409)

Metric	Baseline (LogReg)	XGBoost (tuned)
PR-AUC	0.2316	0.2503
ROC-AUC	0.8148	0.8043
Recall @ top-10%	0.3763	0.4194
Lift @ top-10%	3.76x	4.19x
Best-F1	0.3254	0.3470
Best-F1 threshold	0.331	0.386

See lift_chart.png.

4. The 3 Actionable Segments

1. **Contract_ord below 0.69** — 42.7% churn vs 6.8% in the other half (6.32x), affecting ~3875 customers.
2. **tenure below 29.00** — 39.5% churn vs 13.2% in the other half (2.98x), affecting ~3569 customers.
3. **charges_per_service above 13.97** — 43.4% churn vs 9.7% in the other half (4.45x), affecting ~3520 customers.

Full breakdown: segments.md. SHAP attribution: shap_beeswarm.png, shap_bar.png.

5. Recommended Actions

1. **Stand up a weekly top-10% risk list** from the model — at current performance, contacting these ~141 customers per scoring run captures **41.9%** of all coming churners.
2. **Convert month-to-month customers** to annual contracts via targeted discounting, prioritising tenure < 12 months.
3. **Reduce electronic-check usage** with a one-time auto-pay enrollment incentive — this single switch correlates strongly with retention.
4. **Investigate fiber-optic churn root cause** with the network/ops team — this segment loses high-ARPU customers at the fastest rate.

Estimated revenue impact

~\$42,818/yr retained, assuming:

- top-10% targeting captures 41.9% of churners,
- 30% save-rate from outreach (industry benchmark; replace with internal data once available),
- average ARPU of saved customers = \$74.44/mo × 12.

Caveats

- Split is time-aware on tenure, not calendar time — production deployment must re-validate on rolling-window holdouts.
- Causation vs correlation: SHAP attributions identify drivers in the model, not real-world levers. Validate via A/B test before rolling discounts at scale.
- The 30% save-rate is a placeholder. Final ROI depends on actual lift from outreach, which we recommend measuring with a randomized holdout group from launch.